

TASK 2

Pandas &

Using matplotlib for
visualization

مش شغال عشان بنددا

Muhammad Elsayed Aly

AI2616



Rod map for the presentation

- 1- Load the Data
- 2- Data Cleaning
- 3- Exploratory Data Analysis (EDA)
- 4- Analysis & Insights
- 5- Conclusion

1-Frist thing frist : Load the Data

Load the data from "sorce"= [excel , json , cvs , sql ,]

In that case we use the excel data called " tanic.csv"

- And here how it's done

```
import pandas as pd  
df= pd.read_csv("tanic.csv")
```

- Now u can access the data that are in the file tanic.csv

second

Clean and adjust the data using pandas

Filter and correct the data (count NaN values and check for duplicates)

Fill missing values (NaN)

Remove the Cabin column for two reasons: the data was largely missing, and it's not critical in this context

Impute missing values with the most frequent value, as it has the highest statistical likelihood of being correct

Visualize the data

3-(EDA) then clean the data

`df.shape` # seeing the shape

`df.isnull().sum()` # counting the NaN data

`df.duplicated().sum()` # counting the duplicated data

After EDA I notice that are the shape = 891 *12

So I deleted the last column = 'Cabin' shape = 891 *11

`df.drop(columns=['Cabin'], inplace=True)` # removing the cabin for 2 reasons

#1 was missing data and not that important in that case

,

Analysis & Insights

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
# filling out NaN (blanked) values with mean of the age
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0]) --> frist chr from the  
'Embarked' colum
```

```
# Filling missing values with the most frequent value
```

```
# since statistically it has the highest probability of being correct
```

Questions that we should ask our self ?

1- % of survived mans , womans

2- what is the % of woman and mans on ?

before and after

3- embarked? --> in py file

4- pclass 1 , 2 ,3 ?

1- Victims vs Rescued

There was 891 ppl on titanic so here we want to know

how many ppl Survived ?

Here using to use matplotlib to visualization

```
29 df['Survived'].value_counts().plot(kind='bar')
30 plt.xticks([0, 1], ['Victims', 'Rescued'], rotation=0)
31 plt.title('Titanic Survivors')
32 plt.grid(axis='y', linestyle='--', alpha=0.7)
33 plt.yticks(range(0, 600, 50))
34 plt.show()
35
```

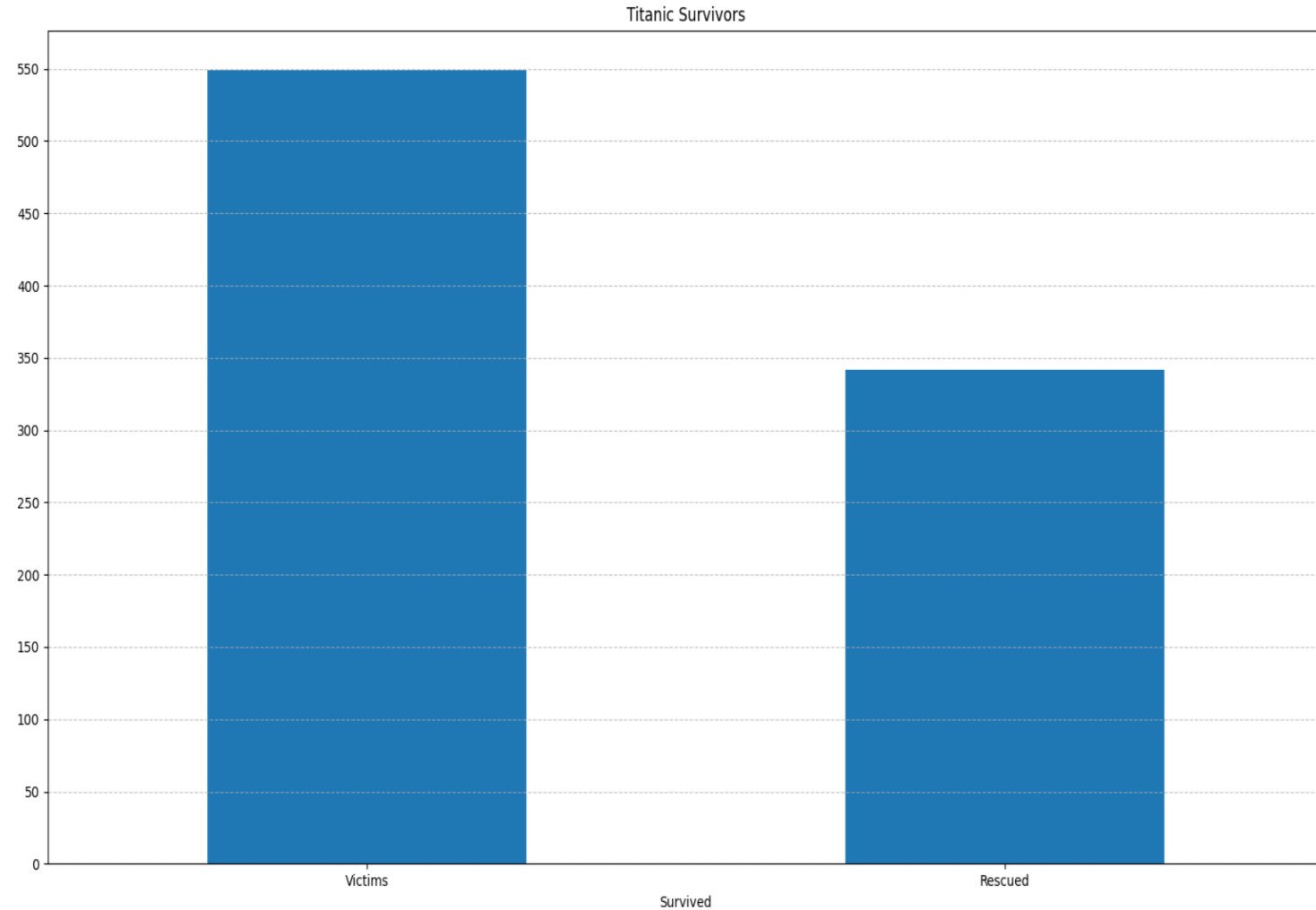

**Given that there theTotal
passanger = 891 person
on titanic**

Rescued = 342 person

Victims = 549 person

The code of the bar is in file named "task 2

what is the % of woman and mans on titanic?

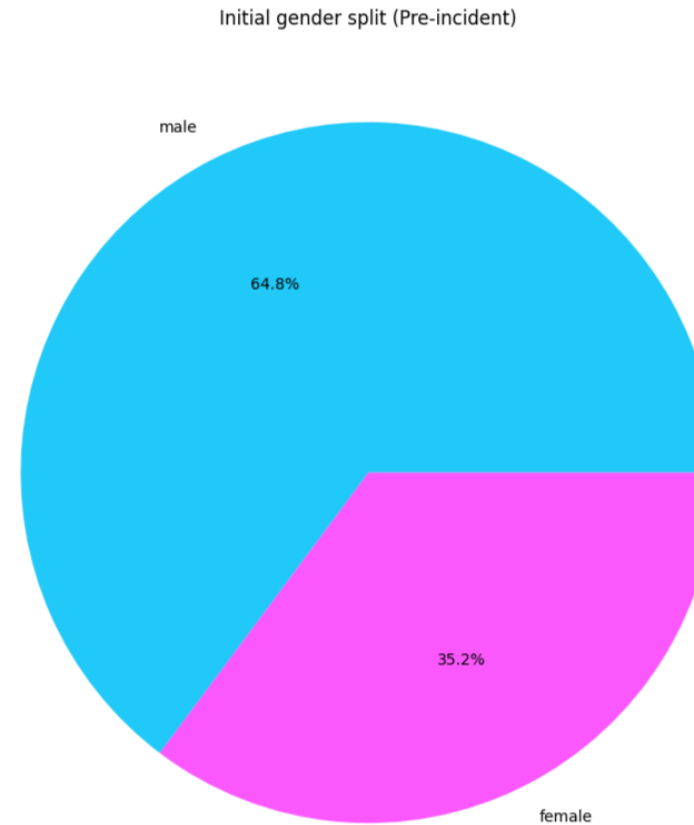


2 - Initial gender split (Pre-incident) vs

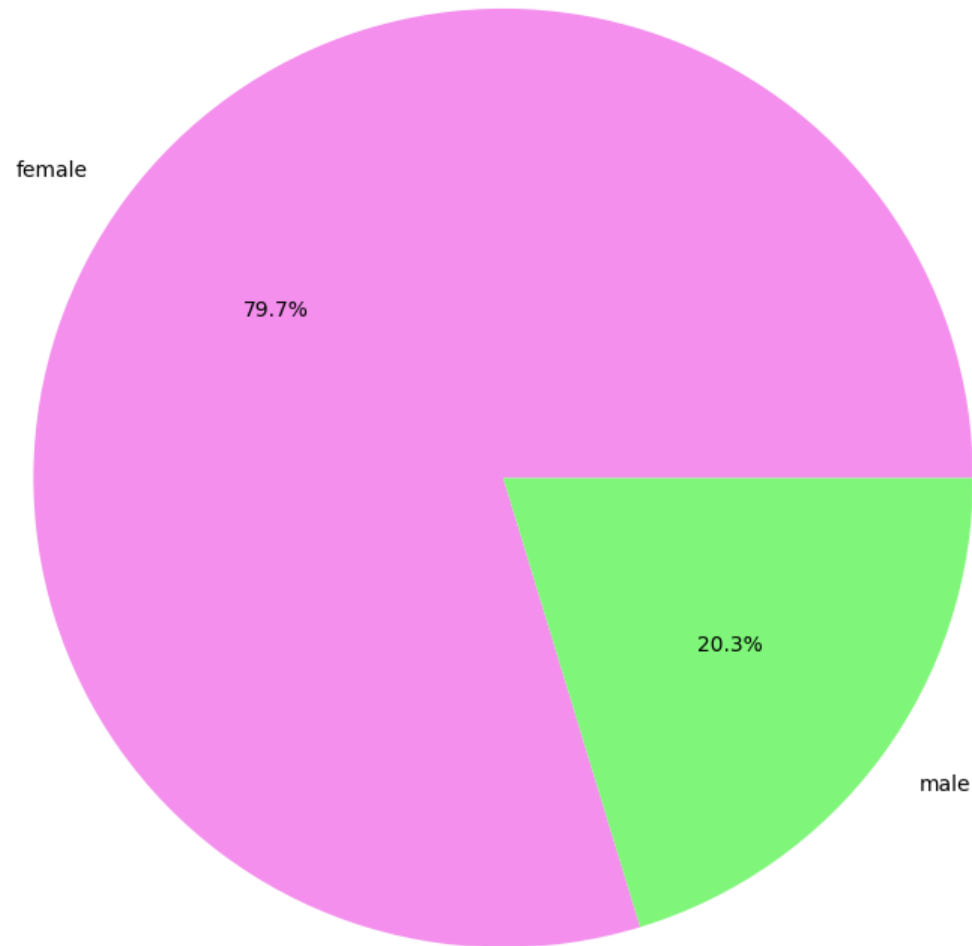
Number of man = 561 person

Number of woman = 330 pers

Before the incident



gender split (After-incident)



gender split (After-incident)

- Number of man = 68 person
- Number of woman = 273 person
- After the incident

Survival rate varies by passenger class

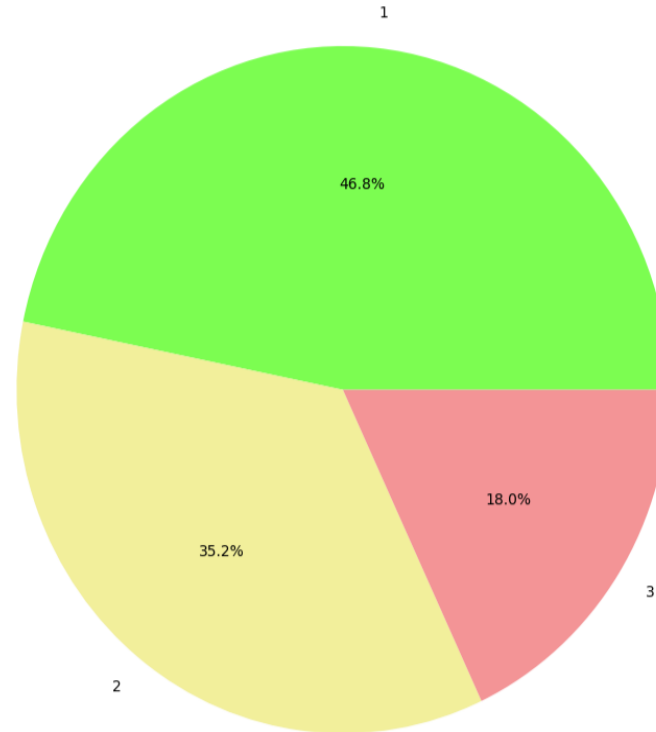
Here we can see that ppl who
was on the frist class have the
highst rate of Surviving Victims

st class = 160

nd class = 121

rd class = 61

Survival rate varies by passenger class



5- Conclusion

- From the pie charts, we can observe that passengers in **Pclass 2 and Pclass 3** had the **lowest survival rates among the three passenger classes**.
- Before the incident, **men represented about 63% of the passengers (561 men)**. After the incident, their proportion **dropped to 20.3%**, which indicates a significant decrease.
- This may also suggest that **a larger proportion of women were located in the higher classes (Pclass 1 and Pclass 2)**.
- On the other hand, **women made up about 35% of the passengers before the incident**. After the incident, their proportion **increased to 79.7%**, which represents a substantial rise in the survivor distribution.